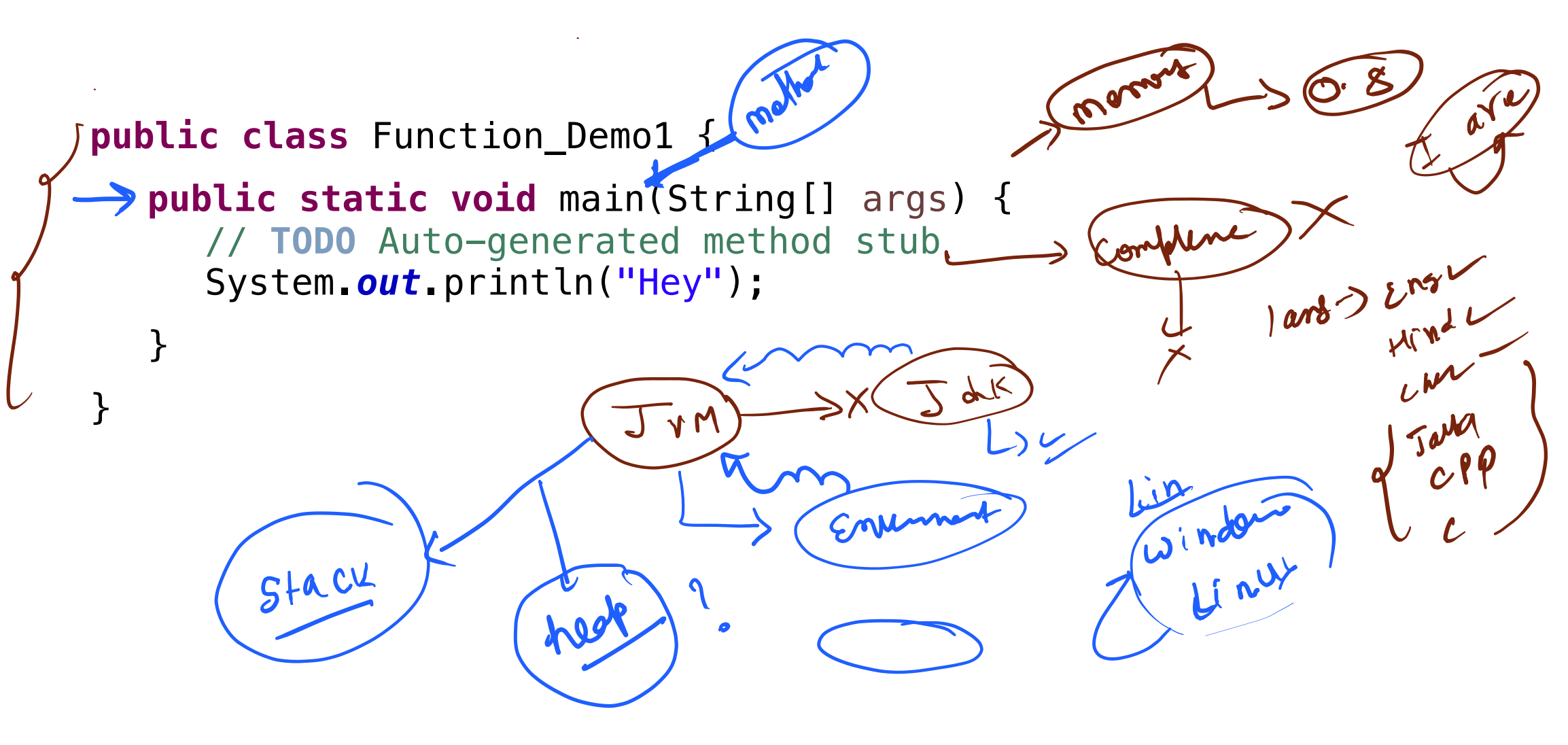
Lec-6

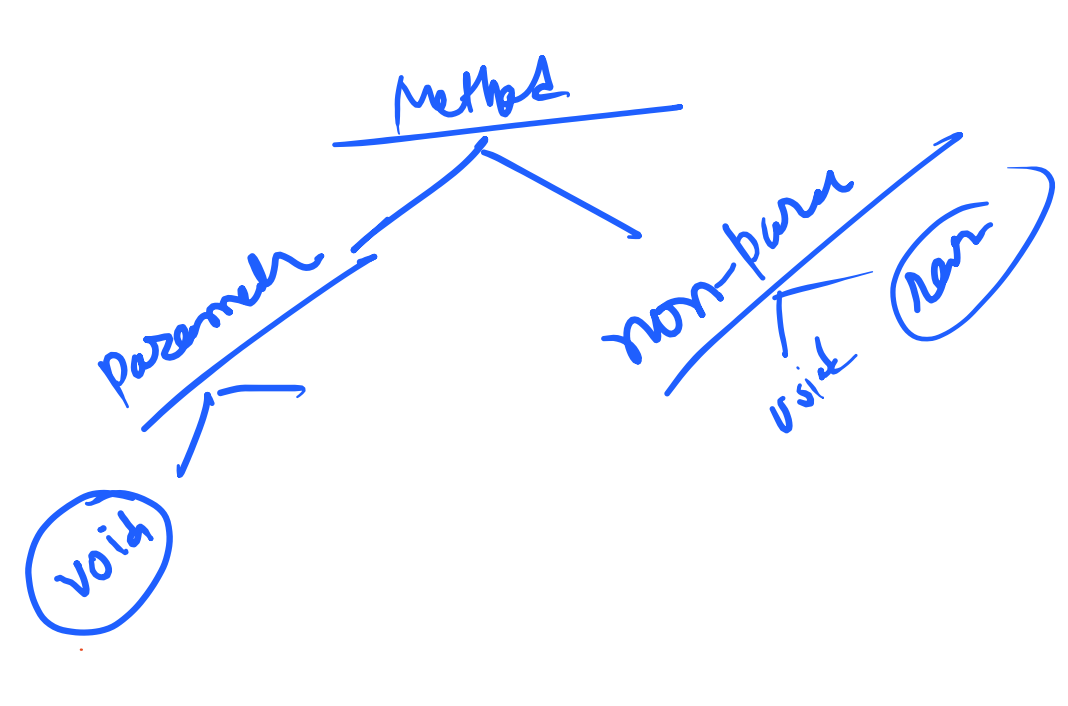
24/10/26 9:57 AM

Function → method

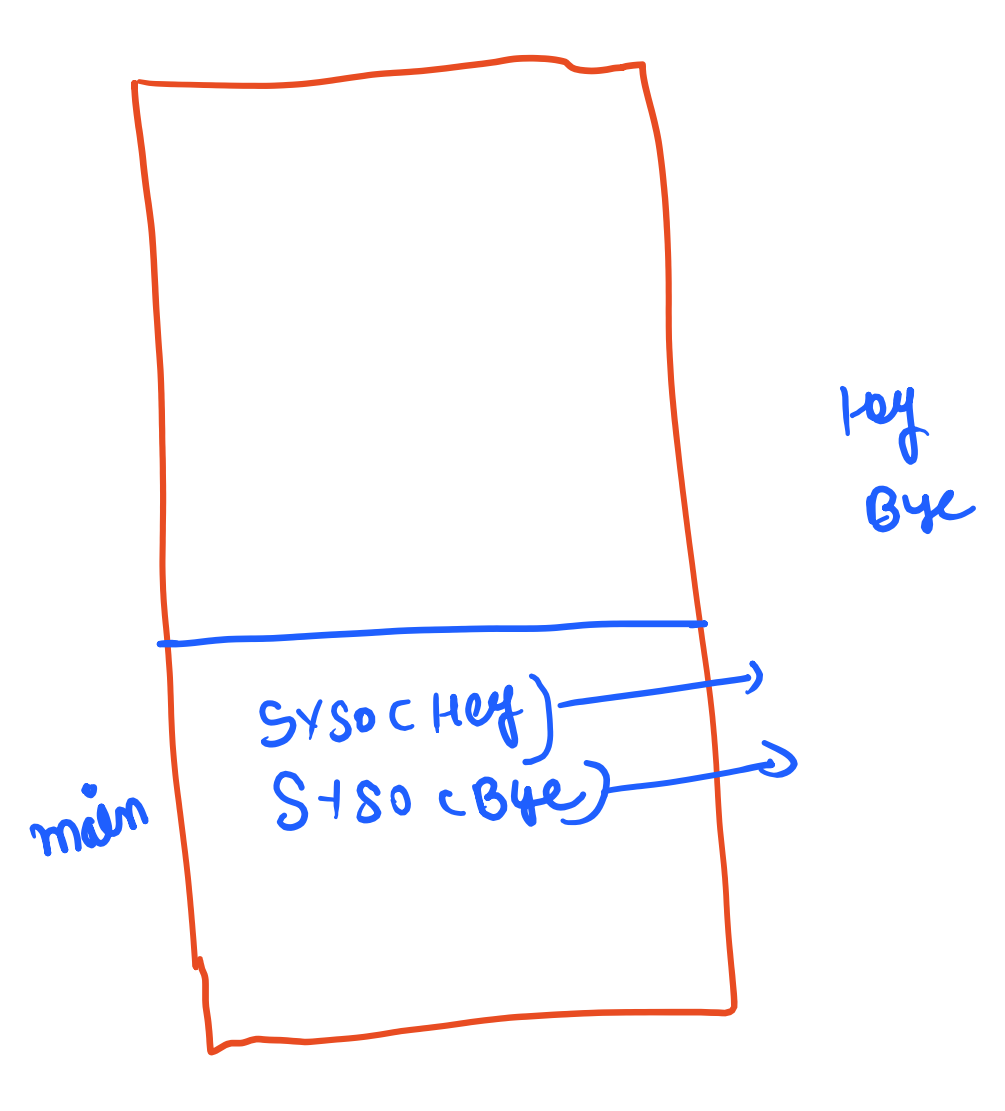
#


method
memory
0.8
I are
Complience
lang → Engl
Hindi
Chi
Java
cpp
c
JVM
JDK
Environment
stack
heap
?
Lin
window
Linux

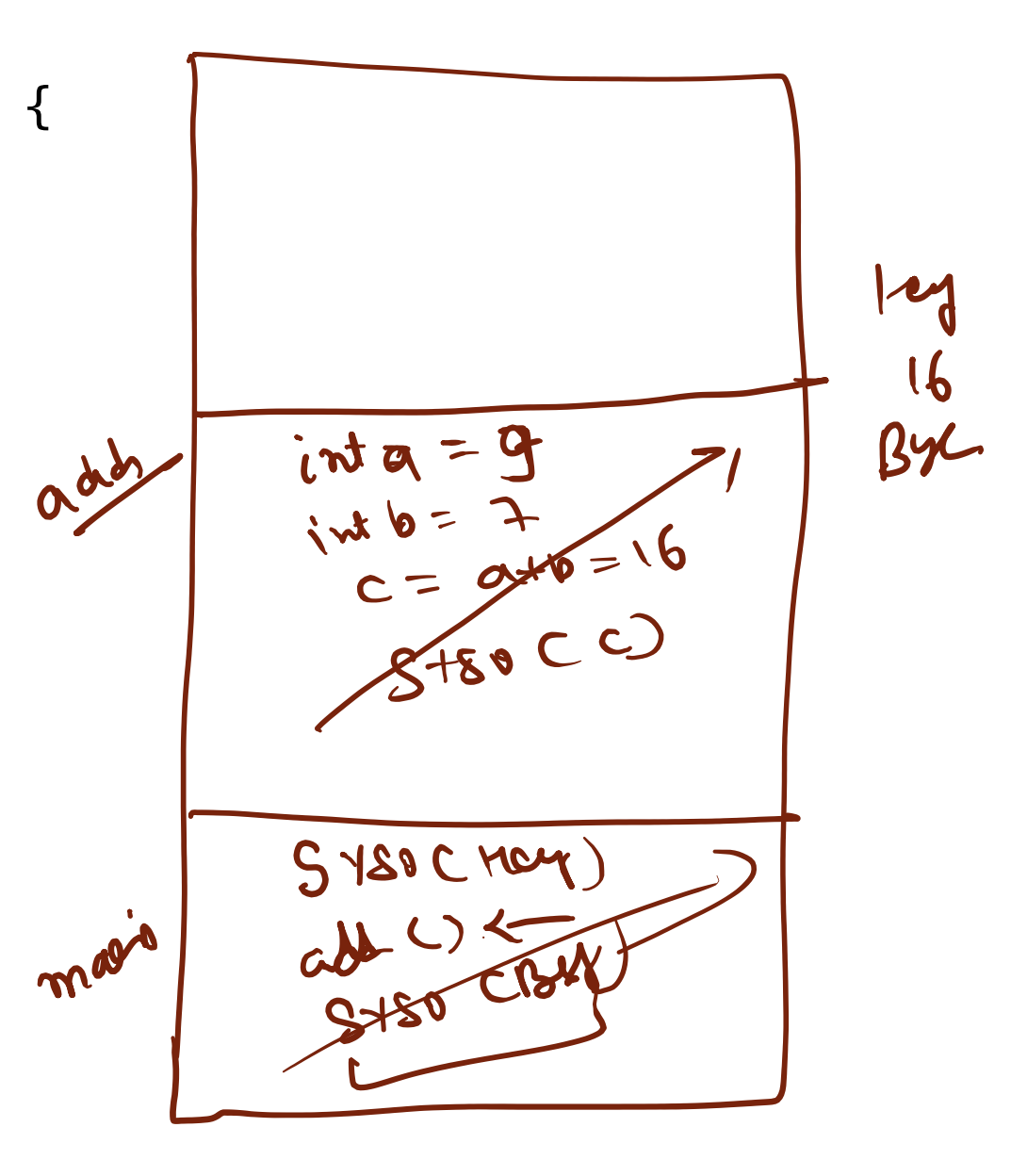
```
public class Function_Demo1 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        System.out.println("Hey");

    }
}
```


Method
parameter
non-para
void
void
ram

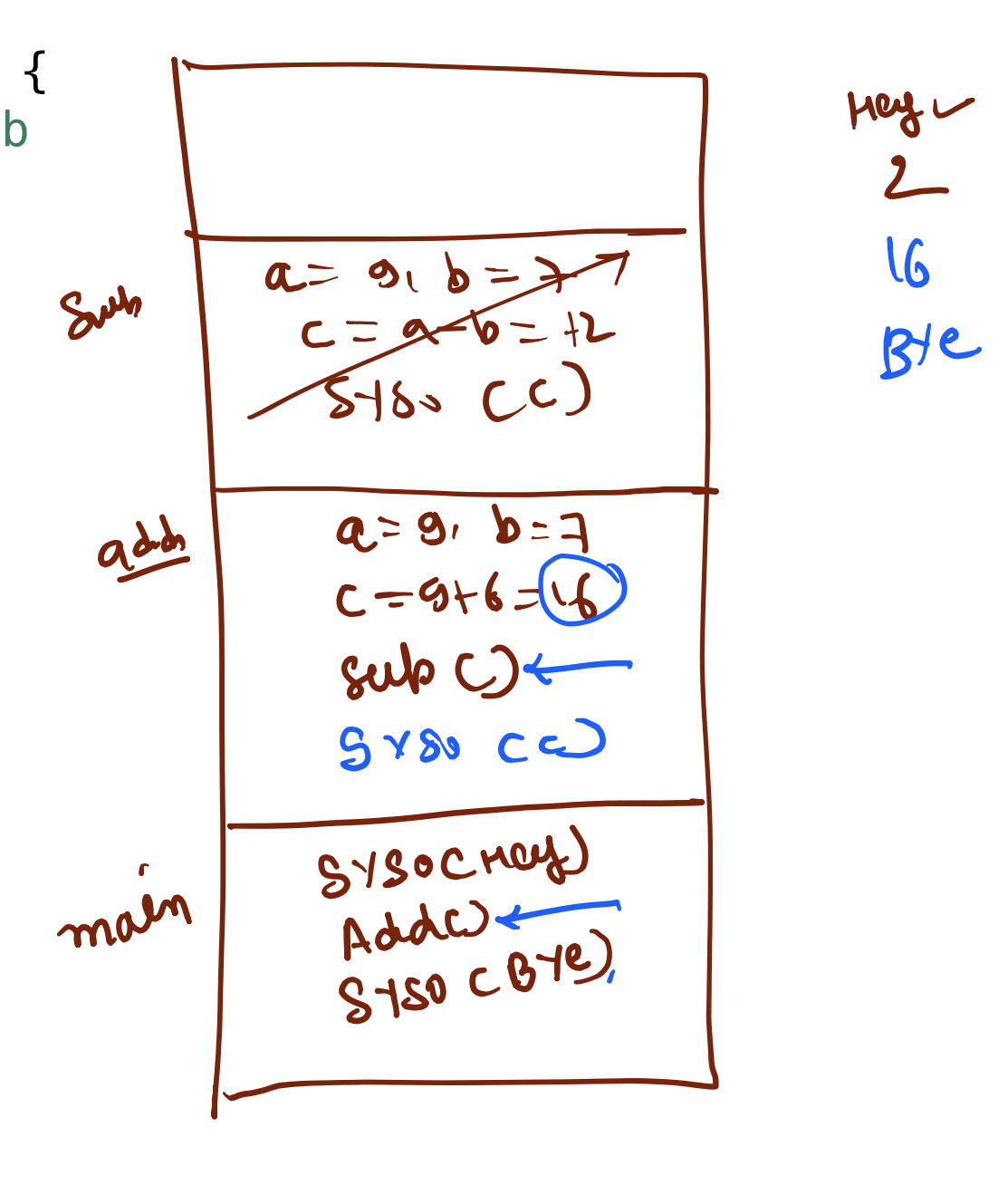
```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        System.out.println("Hey");
        System.out.println("Bye");
    }

    public static void add() {
        int a = 9;
        int b = 7;
        int c = a + b;
        System.out.println(c);
    }
```


Hey
Bye
main
Sysoc(Hey)
Sysoc(Bye)

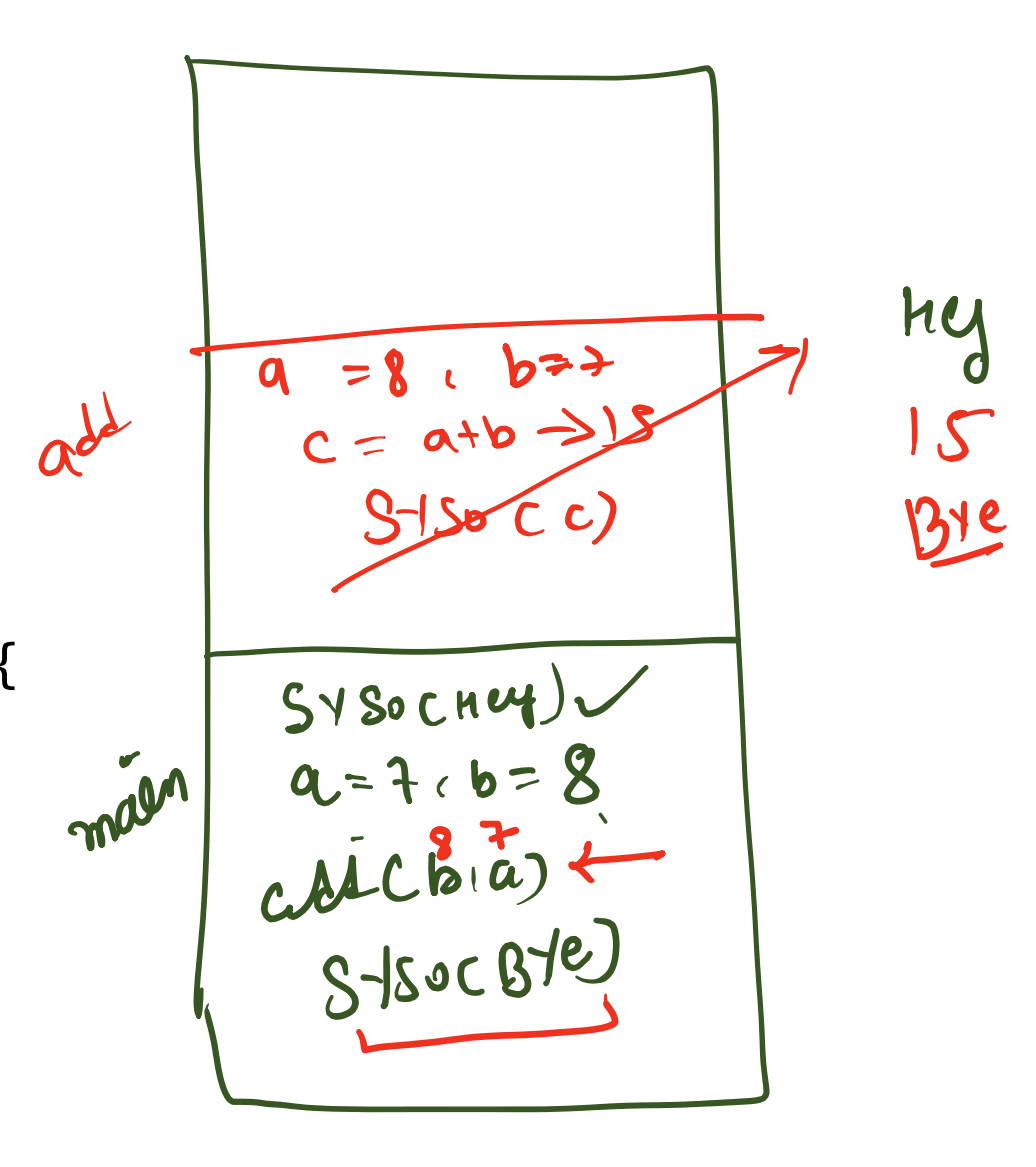
```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        System.out.println("Hey");
        add();
        System.out.println("Bye");
    }

    public static void add() {
        int a = 9;
        int b = 7;
        int c = a + b;
        System.out.println(c);
    }
```


Hey
16
Bye
add
int a = 9
int b = 7
c = a+b = 16
Syso(c)
main
Syso(Hey)
add()
Syso(Bye)

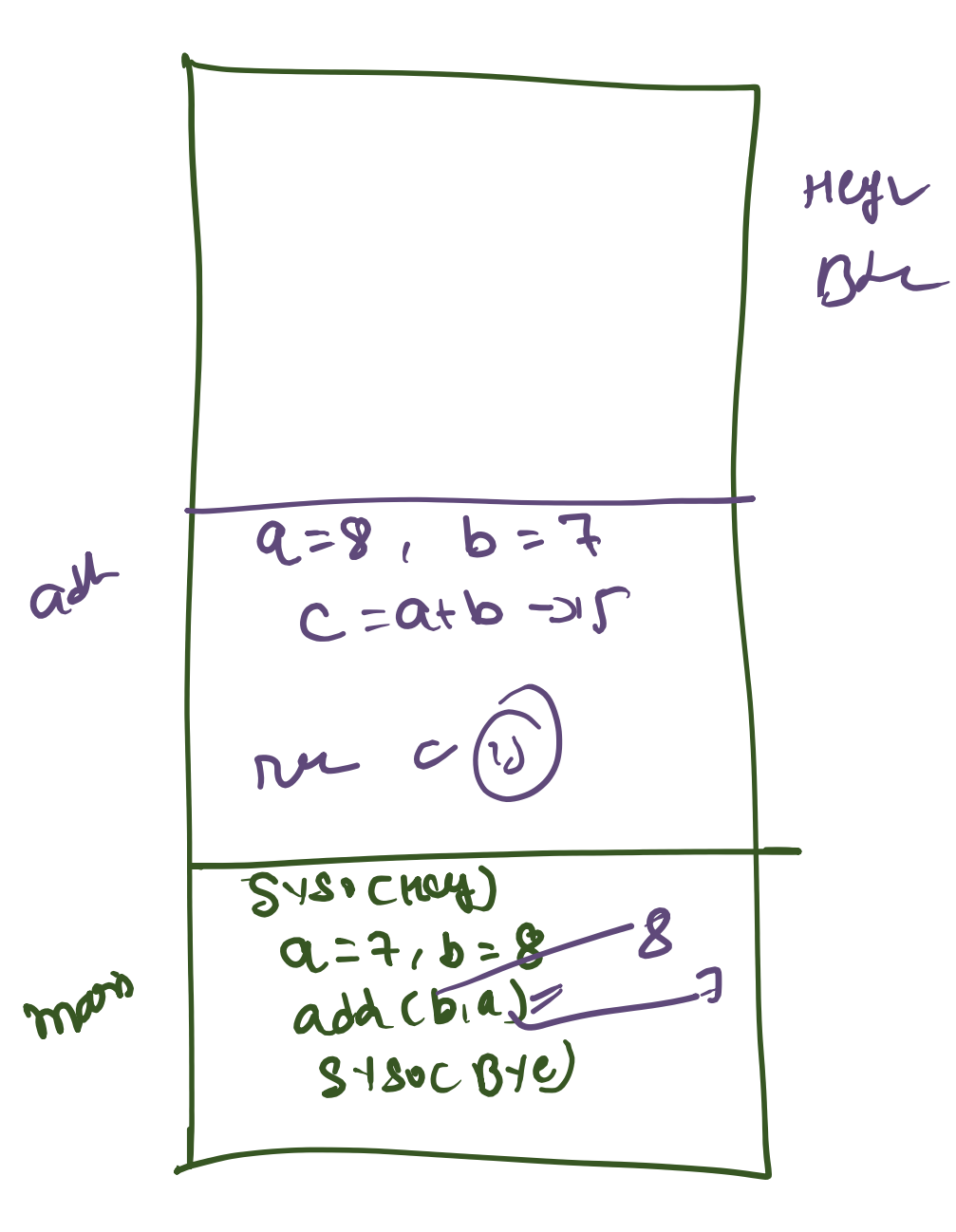
```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        System.out.println("Hey");
        add();
        System.out.println("Bye");
    }

    public static void add() {
        int a = 9;
        int b = 7;
        int c = a + b;
        sub();
        System.out.println(c);
    }

    public static void sub() {
        int a = 9;
        int b = 7;
        int c = a - b;
        System.out.println(c);
    }
```


local
local
Hey
2
16
Bye
Sub
a = 9, b = 7
c = a-b = 2
Syso(c)
add
a = 9, b = 7
c = 9+6 = 16
sub()
Syso(c)
main
Sysoc(Hey)
Add()
Syso(Bye)

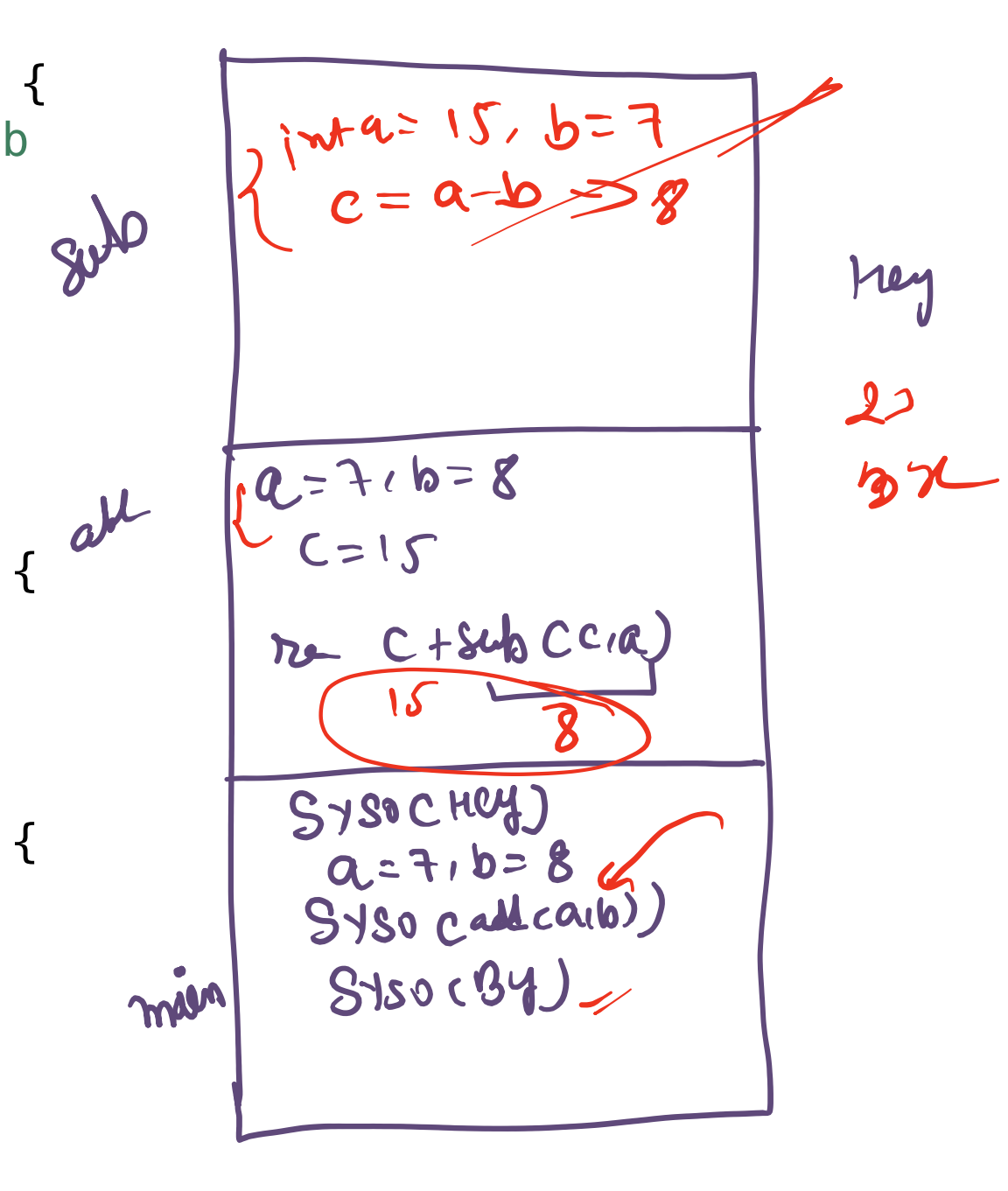
```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        System.out.println("Hey");
        int a = 7;
        int b = 8;
        add(b, a);
        System.out.println("Bye");
    }

    public static void add(int a, int b) {
        int c = a + b;
        System.out.println(c);
    }
```


content
Hey
15
Bye
add
a = 8, b = 7
c = a+b → 15
Syso(c)
main
Sysoc(Hey)
a = 7, b = 8
add(b,a)
Sysoc(Bye)

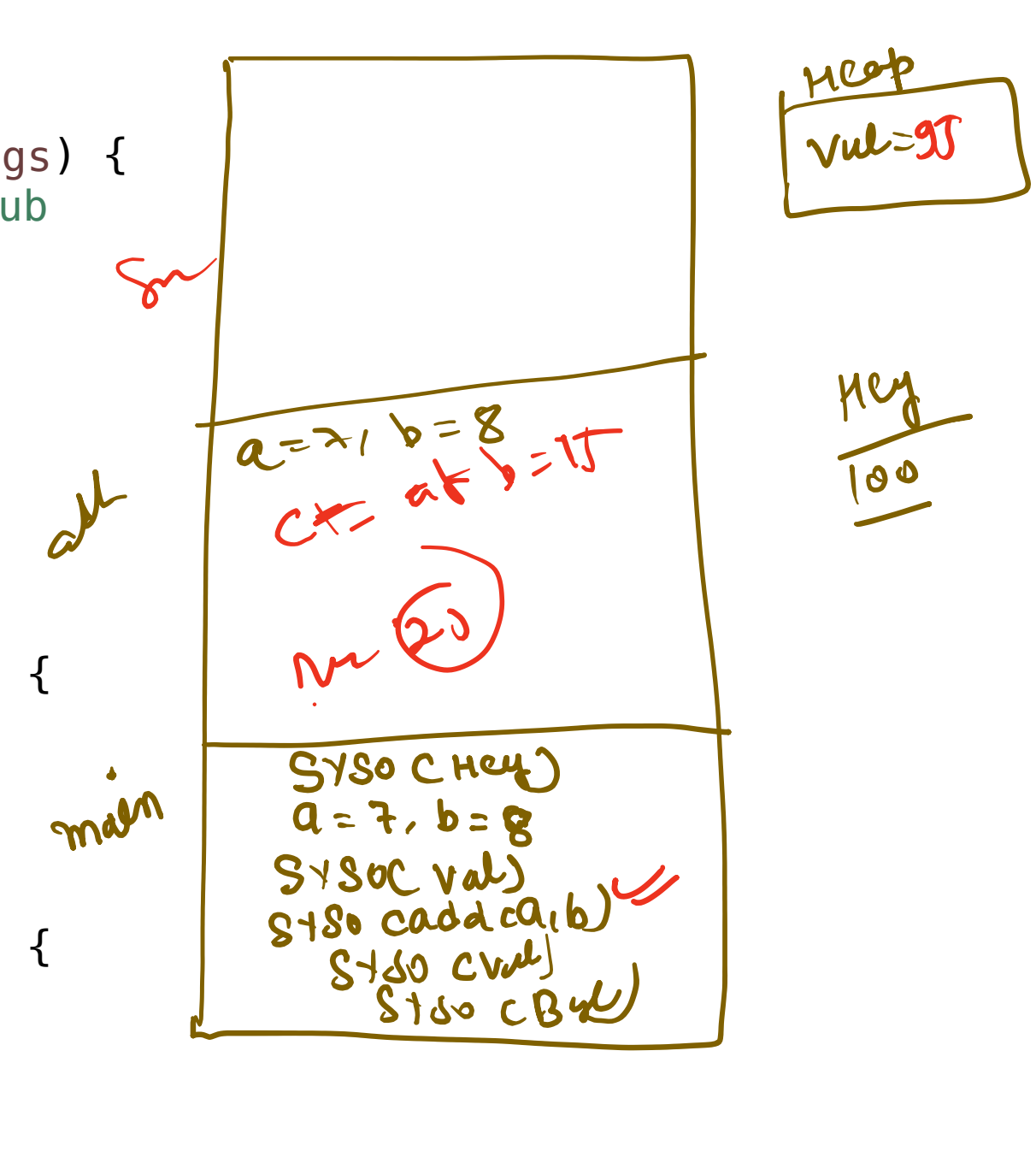
```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        System.out.println("Hey");
        int a = 7;
        int b = 8;
        add(b, a);
        System.out.println("Bye");
    }

    public static int add(int a, int b) {
        int c = a + b;
        return c;
    }
```


15
Hey
Bye
add
a = 8, b = 7
c = a+b → 15
re c 15
main
Syso(Hey)
a = 7, b = 8
add(b,a)
Sysoc(Bye)

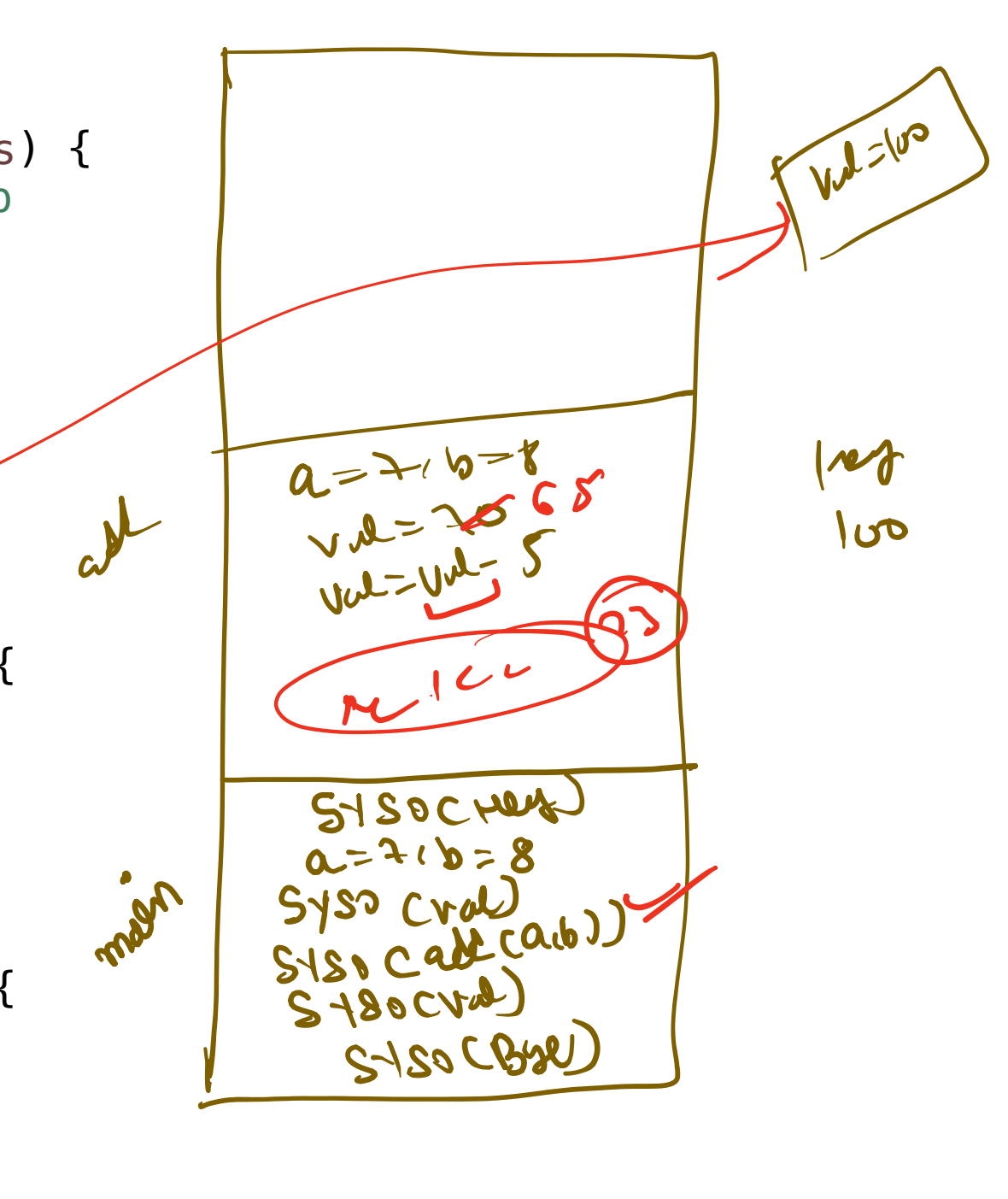
```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        System.out.println("Hey");
        int a = 7;
        int b = 8;
        System.out.println(add(a, b));
        System.out.println("Bye");
    }

    public static int add(int a, int b) {
        int c = a + b;
        return c + sub(c, a);
    }

    public static int sub(int a, int b) {
        int c = a - b;
        return c;
    }
```

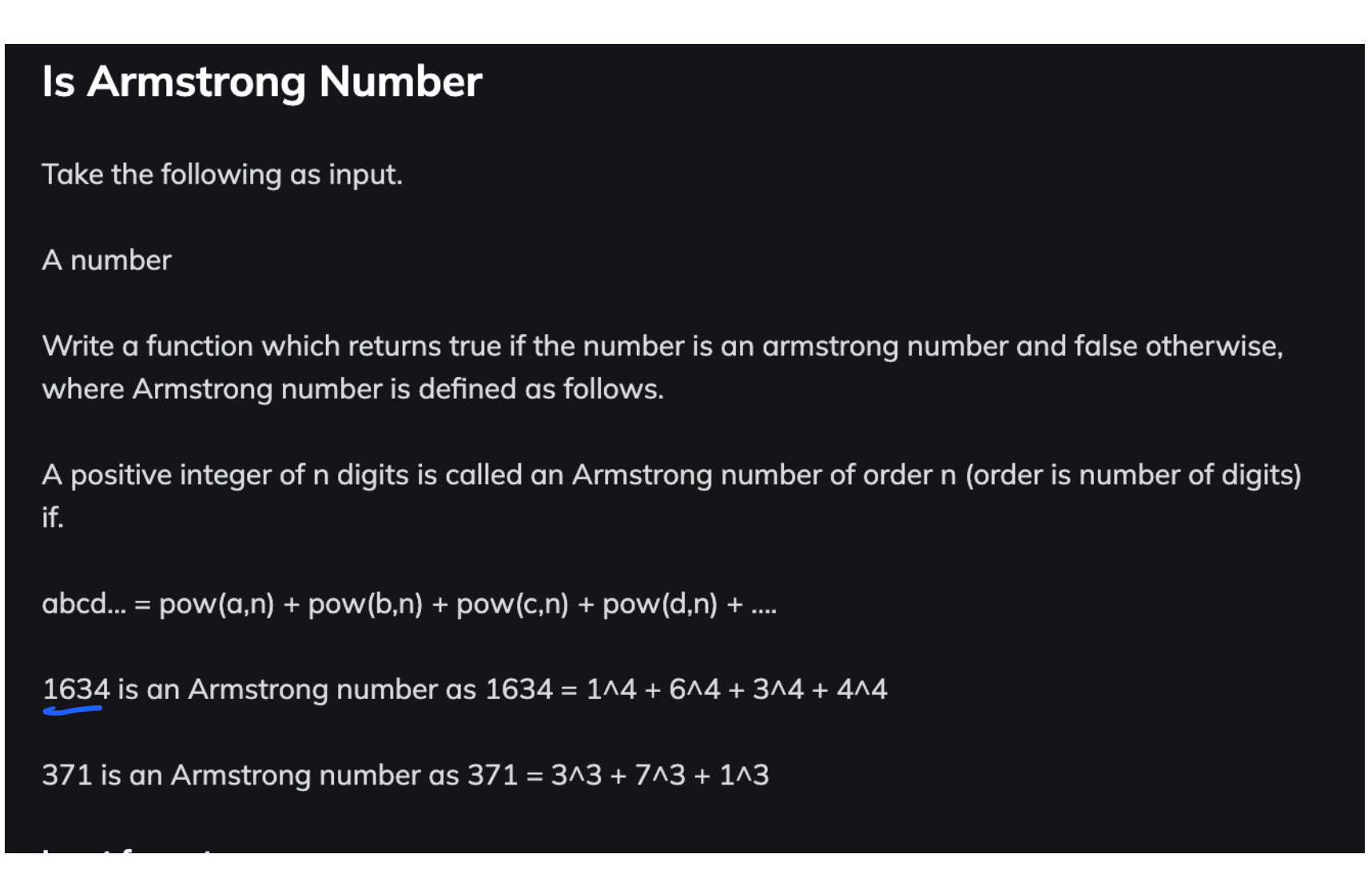

15 7
sub
int a = 15, b = 7
c = a-b → 8
Hey
23
Bye
add
a = 7, b = 8
c = 15
re c+sub(c,a)
15 8
main
Sysoc(Hey)
a = 7, b = 8
Syso(add(a,b))
Syso(Bye)

```
static int val = 100;

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        System.out.println("Hey");
        int a = 7;
        int b = 8;
        System.out.println(val);// 100
        System.out.println(add(a, b));
        System.out.println(val);// ??
        System.out.println("Bye");
    }

    public static int add(int a, int b) {
        val = val - 5;
        int c = a + b;
        return c + sub(c, a);
    }

    public static int sub(int a, int b) {
        int c = a - b;
        return c;
    }
```


Heap
val = 95
Sn
Hey
100
add
a = 7, b = 8
c = a+b = 15
re 23
main
Syso(Hey)
a = 7, b = 8
Sysoc(val)
Syso(add(a,b))
Syso(val)
Syso(Bye)

```
static int val = 100;

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        System.out.println("Hey");
        int a = 7;
        int b = 8;
        System.out.println(val);// 100
        System.out.println(add(a, b));
        System.out.println(val);// ??
        System.out.println("Bye");
    }

    public static int add(int a, int b) {
        int val = 70;
        val = val - 5;
        int c = a + b;
        return c + sub(c, a);
    }

    public static int sub(int a, int b) {
        int c = a - b;
        return c;
    }
```


local
val = 100
Hey
100
add
a = 7, b = 8
val = 65
val = val - 5
re 100
23
main
Sysoc(Hey)
a = 7, b = 8
Syso(val)
Syso(add(a,b))
Syso(val)
Syso(Bye)



Is Armstrong Number

Take the following as input.

A number

Write a function which returns true if the number is an armstrong number and false otherwise, where Armstrong number is defined as follows.

A positive integer of n digits is called an Armstrong number of order n (order is number of digits) if.

abcd... = pow(a,n) + pow(b,n) + pow(c,n) + pow(d,n) + ...

1634 is an Armstrong number as 1634 = 1^4 + 6^4 + 3^4 + 4^4

371 is an Armstrong number as 371 = 3^3 + 7^3 + 1^3

N = 371

1634

$1^4+6^4+3^4+4^4$

1634

4 3 6 1

n = 1634 → 163 → 16 → 1 → 0

while(n>0){
n=n/10
c++
}

1634 = 1+6+3+4

rem = n%10

sum = sum+rem

n = n/10

1634 → cod = 4

sum = $4^4 + 3^4 + 6^4 + 1^4$

163

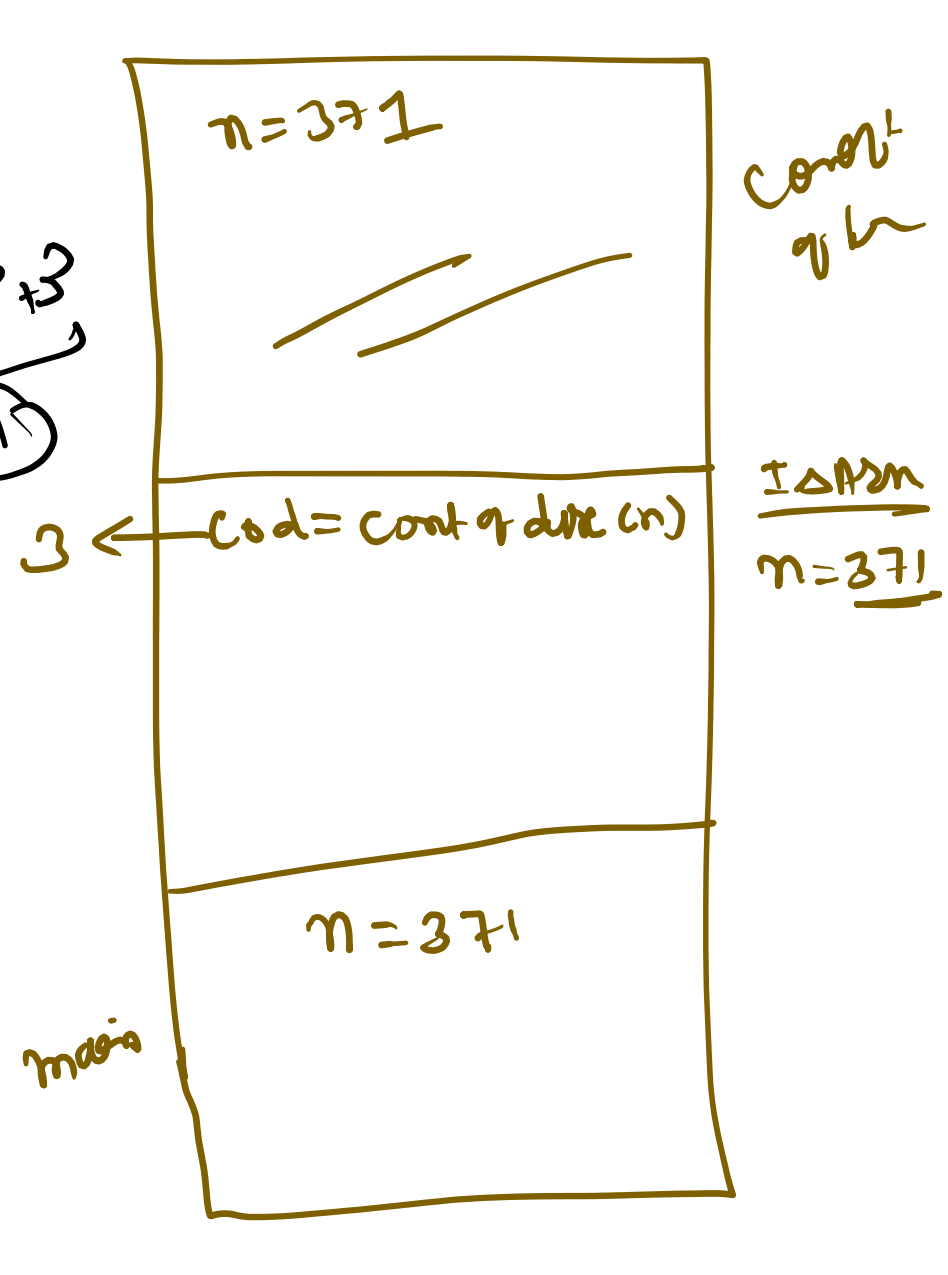
16

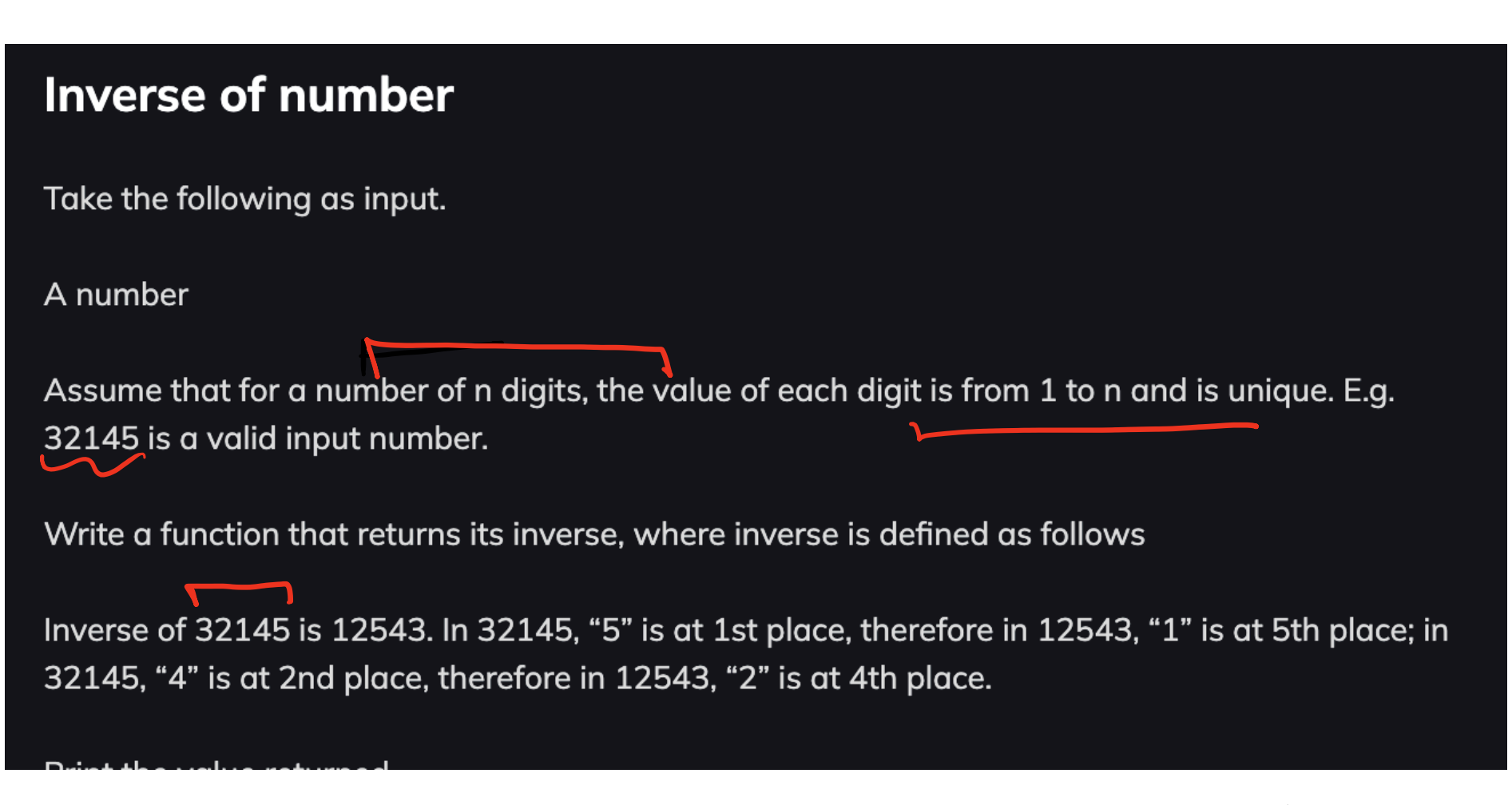
1

0

while(n>0){
rem = n%10
sum = sum + rem^cod
n = n/10
}

```
import java.util.Scanner;

public class Is_Armstrong_Number {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        System.out.println(Is_Armstrong(n));
    }
    public static boolean Is_Armstrong(int n) {
        int cod = count_of_digit(n);// 4
        int sum=0;
        while(n>0) {
            int rem=n%10;
            sum= (int) (sum + Math.pow(rem, cod));
            n=n/10;
        }
        if(sum==n) {
            return true;
        }
        else {
            return false;
        }
    }
    public static int count_of_digit(int n) {
        int c = 0;
        while (n > 0) {
            c++;
            n = n / 10;
        }
        return c;
    }

}
```


sum = $1^3+7^3+3^3$
371
rem = n%10 → 1
371>0
37
3
0
n = 371
count
3
cod = count_of_digit(n)
IsArm
n = 371
main
n = 371



Inverse of number

Take the following as input.

A number

Assume that for a number of n digits, the value of each digit is from 1 to n and is unique. E.g. 32145 is a valid input number.

Write a function that returns its inverse, where inverse is defined as follows

Inverse of 32145 is 12543. In 32145, “5” is at 1st place, therefore in 12543, “1” is at 5th place; in 32145, “4” is at 2nd place, therefore in 12543, “2” is at 4th place.

N = 32145
5 4 3 2 1

1 2 5 4 3
5 4 3 2 1

N = 32145

3 2 1 6 4 5
6 5 4 3 2 1

3 1 2 6 5 4
6 5 4 3 2 1

$32145 = 3\times10^4 + 2\times10^3 + 1\times10^2 + 4\times10^1 + 5\times10^0$

32145
5 4 3 2 1

$= 1\times10^{5-1} + 2\times10^{4-1} + 3\times10^{1-1} + 4\times10^{2-1} + 5\times10^{3-1}$

p × 10^rem-1

= 10000
2000
3
40
500

12543

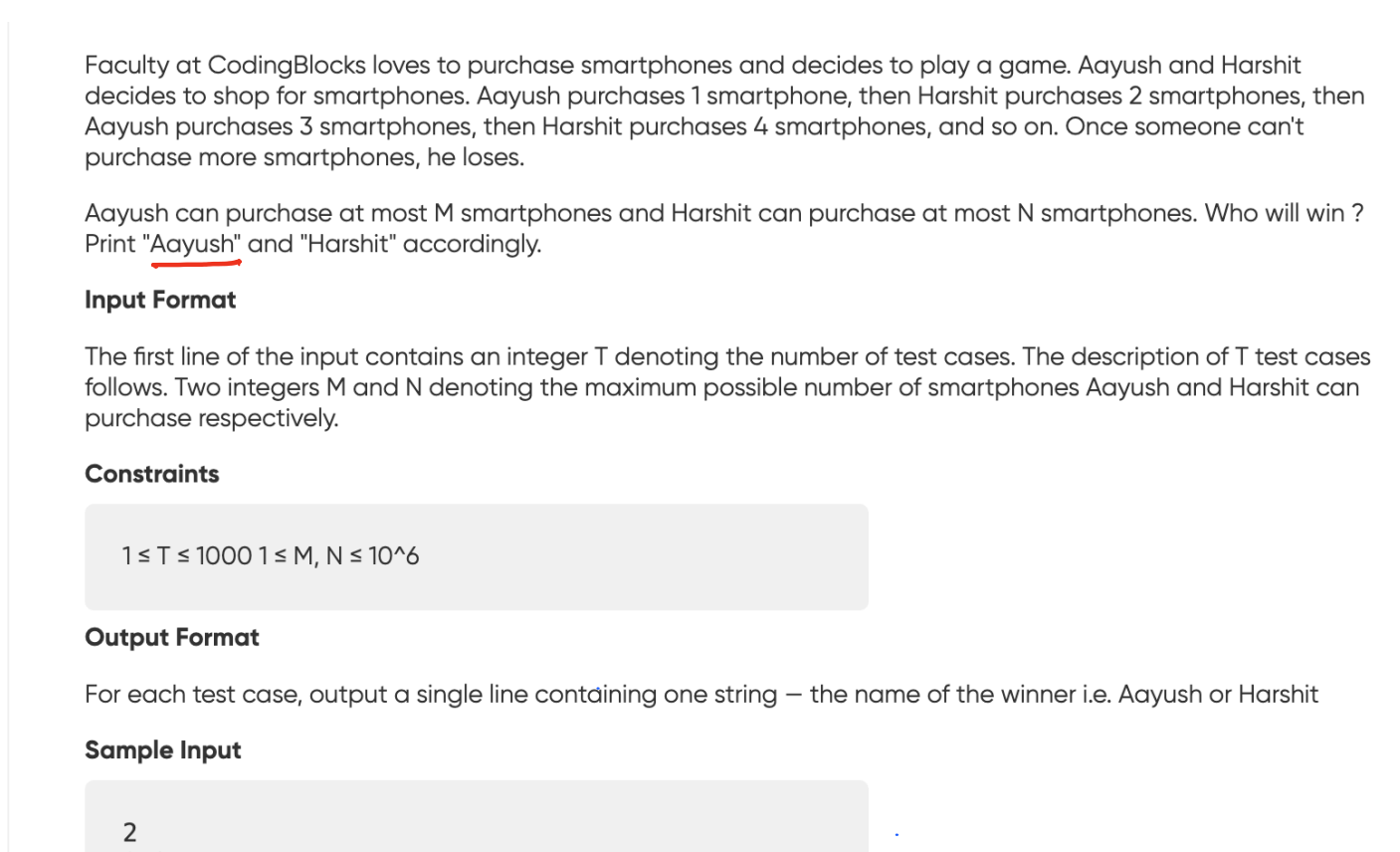
plac = 1 2 3 4 5 6

sum = $1\times10^{5-1} + 2\times10^{4-1} + 3\times10^{1-1} + 4\times10^{2-1} + 5\times10^{3-1}$

32145
5 4 3 2 1

3214 → 2
321
32
3
0

while(n>0){
rem = n%10
sum = sum + place × 10^rem-1
n = n/10
place
}



Faculty at CodingBlocks loves to purchase smartphones and decides to play a game. Aayush and Harshit decides to shop for smartphones. Aayush purchases 1 smartphone, then Harshit purchases 2 smartphones, then Aayush purchases 3 smartphones, then Harshit purchases 4 smartphones, and so on. Once someone can't purchase more smartphones, he loses.

Aayush can purchase at most M smartphones and Harshit can purchase at most N smartphones. Who will win ? Print "Aayush" and "Harshit" accordingly.

Input Format

The first line of the input contains an integer T denoting the number of test cases. The description of T test cases follows. Two integers M and N denoting the maximum possible number of smartphones Aayush and Harshit can purchase respectively.

Constraints

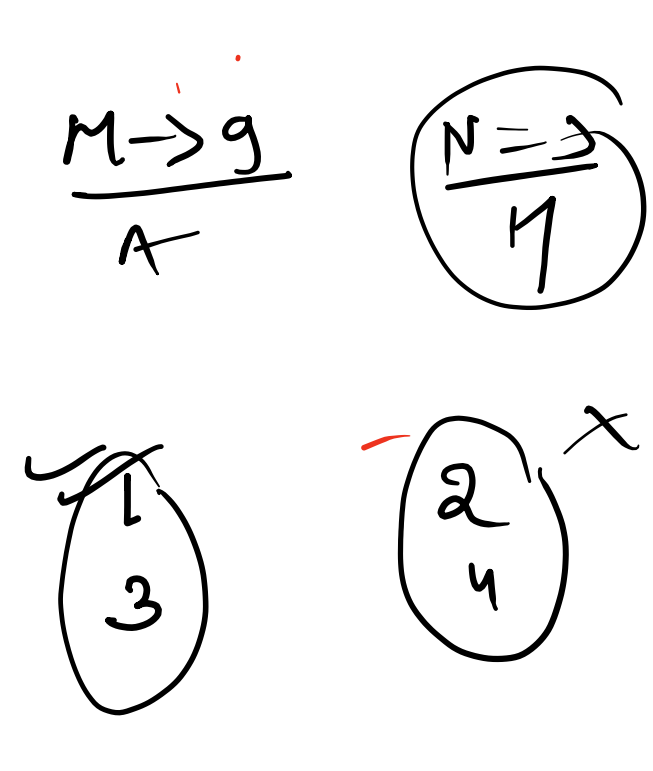
```
1 ≤ T ≤ 1000 1 ≤ M, N ≤ 10^6
```

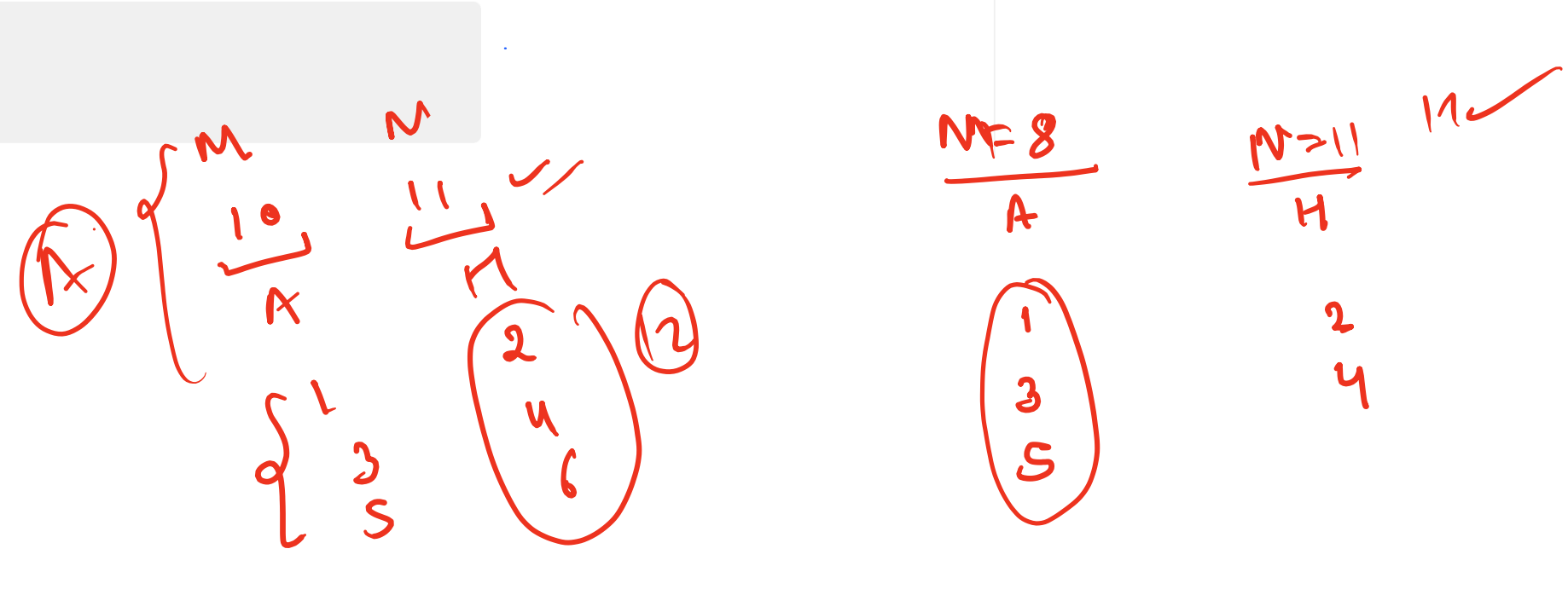
Output Format

For each test case, output a single line containing one string — the name of the winner i.e. Aayush or Harshit

Sample Input

```
2
9 3
8 11
```


M → 9
A
N = 3
H
1
3
2
4


A
M
10
A
N
11
H
1
3
5
2
4
6
12
M = 8
A
1
3
5
N = 11
H
2
4
11

m = 10

a = 1 + 3 + 5

H = 2 + 4 + 6

int a = 0 h = 0

n = 4

4 F 6

phone = 1

while(true){
a = a + phone
if(a > m){
Sysoc(Harshit)
break
}
phone++
H = H + phone
if(H > n){
Syso(Aayush)
break
}
phone++
}